

# A Gaussian Hidden Markov Model for Market-Regime Detection and Regime-Aware Position Sizing

Mathematical specification of the `lng_desk.hmm` module

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## Abstract

We give a complete, self-contained mathematical description of a three-state Gaussian hidden Markov model (HMM) used to classify daily asset returns into latent *bear*, *neutral* and *bull* regimes, and to convert the inferred regime belief into a trading position. The document is written for a reader at the level of a final-year undergraduate in mathematics: we assume familiarity with probability, linear algebra, and multivariable calculus, but we derive every algorithm — the forward–backward recursions, the Baum–Welch (EM) parameter estimator, the Viterbi decoder, online filtering, and the predictive moments — from first principles. Each modelling choice in the implementation is stated explicitly and justified. The notation maps one-to-one onto the code in `src/lng_desk/hmm/`; a symbol dictionary is given in Section 11.

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# 1 Motivation and overview

Daily returns of a traded asset are notoriously hard to model with a single fixed distribution. Empirically, financial markets alternate between qualitatively different *regimes*: calm trending periods with small positive drift and low volatility; directionless choppy periods; and stressed periods with negative drift and large volatility. These regimes are not labelled in the data — we observe only the returns. This is precisely the setting of a *hidden Markov model*: an unobserved (“hidden”) discrete state evolves as a Markov chain, and at each time the state emits an observation through a state-dependent probability law.

We posit  $K = 3$  hidden states with the interpretation

$$\text{state 1} = \text{bear}, \quad \text{state 2} = \text{neutral}, \quad \text{state 3} = \text{bull},$$

and we assume that, *conditional on the current state*, the daily return is Gaussian with a state-specific mean and variance. The bear state carries a negative mean and (typically) the largest variance; the neutral state a near-zero mean and small variance; the bull state a positive mean.

The work splits into four classical problems, which we treat in turn:

- (I) **Evaluation.** Given parameters, compute the likelihood of an observed return series. Solved by the *forward* recursion (Section 3).
- (II) **Learning.** Estimate the parameters from data by maximum likelihood. Solved by the *Baum–Welch* algorithm, an instance of Expectation–Maximisation (Section 4).
- (III) **Decoding.** Recover the most likely sequence of hidden states. Solved by the *Viterbi* algorithm (Section 6).

- (IV) **Prediction & control.** Online, using only past data, compute the current regime belief, predict tomorrow's return distribution, and map it to a position (Sections 7–8).

## 2 The probabilistic model

### 2.1 Random variables and indices

Time is discrete,  $t = 1, 2, \dots, T$ , indexing trading days.<sup>1</sup> At each  $t$  there is

- a **hidden state**  $S_t \in \{1, \dots, K\}$ , with  $K = 3$ ; and
- an **observation**  $r_t \in \mathbb{R}$ , the day's return (e.g. the log return  $r_t = \log P_t - \log P_{t-1}$  of the underlying).

We write  $r_{1:t} := (r_1, \dots, r_t)$  for a prefix of the observation sequence, and similarly  $S_{1:t}$ .

### 2.2 Defining assumptions

The model is characterised by two conditional-independence assumptions.

**Definition 1** (Markov property of the hidden chain). The state sequence  $(S_t)$  is a first-order, time-homogeneous Markov chain:

$$P(S_t | S_{1:t-1}) = P(S_t | S_{t-1}), \quad t \geq 2, \quad (1)$$

and this conditional law does not depend on  $t$ .

**Definition 2** (Conditional independence of emissions). Given the entire state path, the observations are mutually independent and each  $r_t$  depends only on the contemporaneous state  $S_t$ :

$$P(r_{1:T} | S_{1:T}) = \prod_{t=1}^T P(r_t | S_t). \quad (2)$$

### 2.3 Parameters

The model is fully specified by the triple  $\theta = (\boldsymbol{\pi}, A, \{(\mu_k, \sigma_k^2)\})$ :

$$\pi_i := P(S_1 = i), \quad \text{initial distribution, } \boldsymbol{\pi} \in \mathbb{R}^K, \sum_i \pi_i = 1; \quad (3)$$

$$A_{ij} := P(S_{t+1} = j | S_t = i), \quad \text{transition matrix, } A \in \mathbb{R}^{K \times K}, \sum_j A_{ij} = 1; \quad (4)$$

$$b_k(r) := \mathcal{N}(r; \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(r-\mu_k)^2}{2\sigma_k^2}\right), \quad \text{emission density of state } k. \quad (5)$$

Here  $A$  is *row-stochastic*: row  $i$  is the probability vector of the next state given current state  $i$ . We collect the emission parameters as  $\mu = (\mu_1, \dots, \mu_K) \in \mathbb{R}^K$  and  $\sigma^2 = (\sigma_1^2, \dots, \sigma_K^2) \in \mathbb{R}_{>0}^K$ .

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<sup>1</sup>The implementation uses 0-based indexing ( $t = 0, \dots, T-1$ ); this is immaterial to the mathematics.

## 2.4 The joint law and the likelihood

Combining the two assumptions, the joint density of a state path and the observations factorises along the chain:

$$P(S_{1:T}, r_{1:T} | \theta) = \pi_{S_1} b_{S_1}(r_1) \prod_{t=2}^T A_{S_{t-1}S_t} b_{S_t}(r_t). \quad (6)$$

The quantity we ultimately want to maximise over  $\theta$  is the (*marginal*) *likelihood* of the observed data, obtained by summing (6) over all  $K^T$  possible state paths:

$$L(\theta) := P(r_{1:T} | \theta) = \sum_{S_{1:T} \in \{1, \dots, K\}^T} \pi_{S_1} b_{S_1}(r_1) \prod_{t=2}^T A_{S_{t-1}S_t} b_{S_t}(r_t). \quad (7)$$

A direct evaluation of (7) is hopeless: the number of paths grows as  $K^T$  (for one year of daily data,  $3^{252} \approx 10^{120}$ ). The key structural insight — exploited by every algorithm below — is that the sum can be reorganised into  $T$  steps of  $O(K^2)$  work each by *dynamic programming*, because the chain only ever “remembers” the most recent state.

## 3 Evaluation: the forward–backward algorithm

### 3.1 Forward variables

**Definition 3** (Forward variable). For each  $t$  and state  $i$  define the joint density of the data seen so far and the current state,

$$\alpha_t(i) := P(r_{1:t}, S_t = i | \theta). \quad (8)$$

**Proposition 1** (Forward recursion). *The forward variables satisfy*

$$\alpha_1(i) = \pi_i b_i(r_1), \quad (9)$$

$$\alpha_t(j) = \left( \sum_{i=1}^K \alpha_{t-1}(i) A_{ij} \right) b_j(r_t), \quad t = 2, \dots, T, \quad (10)$$

and the likelihood is recovered by marginalising the last step,

$$L(\theta) = \sum_{i=1}^K \alpha_T(i). \quad (11)$$

*Proof.* The initialisation (9) is the definition at  $t = 1$  using  $P(r_1, S_1 = i) = P(S_1 = i)P(r_1 | S_1 = i)$ . For the recursion, condition on the previous state and use the law of total probability:

$$\begin{aligned} \alpha_t(j) &= P(r_{1:t}, S_t = j) = \sum_i P(r_{1:t}, S_{t-1} = i, S_t = j) \\ &= \sum_i P(r_{1:t-1}, S_{t-1} = i) P(S_t = j | S_{t-1} = i) P(r_t | S_t = j), \end{aligned}$$

where the factorisation uses the Markov property (the transition to  $j$  depends only on  $i$ ) and conditional independence of emissions ( $r_t$  depends only on  $S_t = j$ , and is independent of  $r_{1:t-1}$  given  $S_t$ ). Recognising the three factors as  $\alpha_{t-1}(i)$ ,  $A_{ij}$  and  $b_j(r_t)$  gives (10). Finally (11) is  $\sum_i P(r_{1:T}, S_T = i) = P(r_{1:T})$ .  $\square$

Each step (10) costs  $O(K^2)$ , so the whole forward pass is  $O(K^2T)$  — linear in the length of the series. This single fact is what makes HMMs tractable.

### 3.2 Backward variables

To learn parameters we also need the *posterior* distribution of the states given *all* the data. This requires a complementary recursion that propagates information backwards in time. The recursions below all rest on a single conditional-independence fact, which we isolate once.

**Lemma 1** (Markov blanket). *Under the two model assumptions of Section 2, the future is independent of the past given the present state: for every  $t$ ,*

$$(r_{t+1:T}, S_{t+1:T}) \perp (r_{1:t-1}, S_{1:t-1}) \text{ given } S_t. \quad (12)$$

In particular  $P(r_{t+1:T} | S_t, r_{1:t}) = P(r_{t+1:T} | S_t)$ .

*Proof.* Fix  $S_t = i$ . The joint factorisation (6) splits as a product of a “past” block involving only  $(S_{1:t}, r_{1:t})$  and a “future” block  $\prod_{u>t} A_{S_{u-1}S_u} b_{S_u}(r_u)$  that, given  $S_t$ , involves only indices  $u > t$ . Two functions of disjoint coordinate blocks that share only the conditioning variable  $S_t$  are conditionally independent given  $S_t$ ; summing the past block over  $S_{1:t-1}$  and the future block over  $S_{t+1:T}$  leaves (12). The displayed consequence is the special case obtained by marginalising the future states.  $\square$

**Definition 4** (Backward variable).

$$\beta_t(i) := P(r_{t+1:T} | S_t = i, \theta), \quad (13)$$

the likelihood of the *future* observations given the present state, with the convention  $\beta_T(i) = 1$  for all  $i$  (there is no future to condition).

**Proposition 2** (Backward recursion).

$$\beta_T(i) = 1, \quad (14)$$

$$\beta_t(i) = \sum_{j=1}^K A_{ij} b_j(r_{t+1}) \beta_{t+1}(j), \quad t = T-1, \dots, 1. \quad (15)$$

*Proof.* Condition on the next state  $j$ :

$$\beta_t(i) = \sum_j P(S_{t+1} = j, r_{t+1}, r_{t+2:T} | S_t = i).$$

Factor the summand using the Markov property and conditional independence:  $P(S_{t+1} = j | S_t = i) = A_{ij}$ , then  $P(r_{t+1} | S_{t+1} = j) = b_j(r_{t+1})$ , and finally  $P(r_{t+2:T} | S_{t+1} = j, S_t = i, r_{t+1}) = P(r_{t+2:T} | S_{t+1} = j) = \beta_{t+1}(j)$  by the Markov-blanket Lemma 1 (the future beyond  $t+1$  is independent of  $S_t$  and  $r_{t+1}$  given  $S_{t+1}$ ). This is (15).  $\square$

### 3.3 Posterior marginals and pairwise posteriors

The forward and backward variables combine to give the two posterior quantities that drive learning. Throughout,  $L = L(\theta) = \sum_i \alpha_T(i)$ .

**Proposition 3** (Smoothed state posterior). *The posterior probability of being in state  $i$  at time  $t$  given all data is*

$$\gamma_t(i) := P(S_t = i | r_{1:T}, \theta) = \frac{\alpha_t(i) \beta_t(i)}{L}. \quad (16)$$

*Proof.* By definition of the forward and backward variables and the Markov-blanket Lemma 1, which gives  $P(r_{t+1:T} | S_t = i, r_{1:t}) = P(r_{t+1:T} | S_t = i)$ ,

$$\alpha_t(i)\beta_t(i) = P(r_{1:t}, S_t = i) P(r_{t+1:T} | S_t = i) = P(r_{1:t}, S_t = i) P(r_{t+1:T} | S_t = i, r_{1:t}) = P(r_{1:T}, S_t = i).$$

Dividing by  $L = P(r_{1:T})$  gives the conditional (16).  $\square$

**Proposition 4** (Pairwise (transition) posterior). *The joint posterior of consecutive states is, for  $t = 1, \dots, T-1$ ,*

$$\xi_t(i, j) := P(S_t = i, S_{t+1} = j | r_{1:T}, \theta) = \frac{\alpha_t(i) A_{ij} b_j(r_{t+1}) \beta_{t+1}(j)}{L}. \quad (17)$$

*Consistency holds:*  $\sum_j \xi_t(i, j) = \gamma_t(i)$ .

*Proof.* Analogous to Proposition 3, splitting the path at the  $t \rightarrow t+1$  transition and using Lemma 1 to detach the future  $r_{t+1:T}$  from the past given  $S_{t+1}$ :  $P(r_{1:T}, S_t = i, S_{t+1} = j) = \alpha_t(i) A_{ij} b_j(r_{t+1}) \beta_{t+1}(j)$ , then normalise by  $L$ .  $\square$

### 3.4 Numerical realisation: the log domain

The forward variables are products of many densities and probabilities; they decay to zero faster than double-precision floating point can represent (underflow) within a few hundred steps. The implementation therefore carries *logarithms* of all quantities and replaces sums of probabilities by the *log-sum-exp* operation.

**Definition 5** (Log-sum-exp). For  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ ,

$$\text{LSE}(x) := \log \sum_{k=1}^n e^{x_k} = m + \log \sum_{k=1}^n e^{x_k - m}, \quad m := \max_k x_k. \quad (18)$$

The right-hand identity is the numerically stable form actually evaluated: after subtracting the maximum  $m$ , the largest exponent is 0, so  $e^{x_k - m} \in (0, 1]$  and no term overflows; at least one term equals 1, so the sum is bounded below by 1 and its logarithm never underflows (individual small terms may still round to 0, harmlessly — it is the *sum* that is protected). With  $\ell\alpha_t(i) := \log \alpha_t(i)$  etc., the forward recursion (10) becomes

$$\ell\alpha_t(j) = \log b_j(r_t) + \text{LSE}_i(\ell\alpha_{t-1}(i) + \log A_{ij}), \quad (19)$$

and the Gaussian log-density is computed directly,

$$\log b_k(r_t) = -\frac{1}{2} \left( \log(2\pi) + \log \sigma_k^2 + \frac{(r_t - \mu_k)^2}{\sigma_k^2} \right), \quad (20)$$

never as the logarithm of an exponentiated quantity. The smoothed posterior in log form reads  $\log \gamma_t(i) = \ell\alpha_t(i) + \ell\beta_t(i) - \ell L$  with  $\ell L = \text{LSE}_i \ell\alpha_T(i)$ .

## 4 Learning: Baum–Welch as Expectation–Maximisation

We now estimate  $\theta$  by (local) maximum likelihood,  $\hat{\theta} = \arg \max_{\theta} \log L(\theta)$ . Because the states are unobserved, the log-likelihood  $\log L(\theta) = \log \sum_{S_{1:T}} P(S_{1:T}, r_{1:T} | \theta)$  contains a logarithm of a sum and has no closed-form maximiser. The Baum–Welch algorithm is the specialisation to HMMs of the Expectation–Maximisation (EM) algorithm, which sidesteps this by iterating two steps that are each tractable.

## 4.1 The EM principle

Let  $s := S_{1:T}$  denote the latent path and  $r := r_{1:T}$  the data. For *any* distribution  $q$  over the latent paths, write the *evidence lower bound* (ELBO)

$$\mathcal{F}(q, \theta) := \sum_s q(s) \log \frac{P(s, r | \theta)}{q(s)}. \quad (21)$$

A one-line manipulation gives the central identity

$$\log L(\theta) = \mathcal{F}(q, \theta) + \text{KL}(q \| P(\cdot | r, \theta)), \quad (22)$$

where  $\text{KL}(q \| p) = \sum_s q(s) \log \frac{q(s)}{p(s)} \geq 0$  is the Kullback–Leibler divergence, equal to zero iff  $q = p$ . Equation (22) holds because  $\text{KL}(q \| P(\cdot | r, \theta)) = \sum_s q(s) \log \frac{q(s)P(r | \theta)}{P(s, r | \theta)} = \log L(\theta) - \mathcal{F}(q, \theta)$ , using  $P(s | r, \theta) = P(s, r | \theta) / P(r | \theta)$ .

EM exploits (22) by coordinate ascent on  $\mathcal{F}$ :

### E-step

At the current estimate  $\theta^{(n)}$ , maximise  $\mathcal{F}$  over  $q$ . Since  $\log L(\theta^{(n)})$  is fixed, (22) is maximised by driving the KL term to zero, i.e. by choosing

$$q^{(n)}(s) = P(s | r, \theta^{(n)}),$$

the smoothed posterior over paths — which the forward–backward algorithm gives us through its marginals  $\gamma, \xi$ .

### M-step

Maximise  $\mathcal{F}(q^{(n)}, \theta)$  over  $\theta$ . Expanding the ELBO (21) as a sum splits it into a  $\theta$ -dependent and a  $\theta$ -independent piece,

$$\mathcal{F}(q, \theta) = \underbrace{\sum_s q(s) \log P(s, r | \theta)}_{Q(\theta | \theta^{(n)})} - \underbrace{\sum_s q(s) \log q(s)}_{H(q), \text{ independent of } \theta}, \quad (23)$$

so maximising  $\mathcal{F}(q^{(n)}, \theta)$  over  $\theta$  is equivalent to maximising the *expected complete-data log-likelihood*

$$Q(\theta | \theta^{(n)}) := \mathbb{E}_{s \sim q^{(n)}}[\log P(s, r | \theta)]. \quad (24)$$

**Proposition 5** (Monotonicity). *The EM iterates never decrease the likelihood:  $\log L(\theta^{(n+1)}) \geq \log L(\theta^{(n)})$ .*

*Proof.* With  $q^{(n)} = P(\cdot | r, \theta^{(n)})$  the KL term in (22) vanishes, so  $\log L(\theta^{(n)}) = \mathcal{F}(q^{(n)}, \theta^{(n)})$ . The M-step gives  $\mathcal{F}(q^{(n)}, \theta^{(n+1)}) \geq \mathcal{F}(q^{(n)}, \theta^{(n)})$ . Finally (22) at  $\theta^{(n+1)}$  with the *same*  $q^{(n)}$  reads  $\log L(\theta^{(n+1)}) = \mathcal{F}(q^{(n)}, \theta^{(n+1)}) + \text{KL}(\dots) \geq \mathcal{F}(q^{(n)}, \theta^{(n+1)}) \geq \log L(\theta^{(n)})$ , the last KL being  $\geq 0$ . Chaining the inequalities proves the claim.  $\square$

Proposition 5 guarantees convergence of the likelihood sequence (it is non-decreasing and, for our bounded-below variance model with a floor, bounded above); it does *not* guarantee convergence to the global maximum. This is why we use multiple restarts (Section 5).

## 4.2 The expected complete-data log-likelihood for an HMM

Taking the logarithm of the factorised joint (6) turns products into sums and exposes the parameters linearly through indicator functions:

$$\log P(s, r | \theta) = \sum_i \mathbf{1}\{S_1 = i\} \log \pi_i + \sum_{t=2}^T \sum_{i,j} \mathbf{1}\{S_{t-1} = i, S_t = j\} \log A_{ij} + \sum_{t=1}^T \sum_i \mathbf{1}\{S_t = i\} \log b_i(r_t). \quad (25)$$

Taking the expectation under  $q^{(n)} = P(\cdot | r, \theta^{(n)})$  and using  $\mathbb{E}[\mathbf{1}\{S_t = i\}] = \gamma_t(i)$  and  $\mathbb{E}[\mathbf{1}\{S_{t-1} = i, S_t = j\}] = \xi_{t-1}(i, j)$  gives a transition term  $\sum_{t=2}^T \xi_{t-1}(i, j) \log A_{ij}$ ; reindexing  $t \mapsto t-1$  so the sum runs over the  $T-1$  transitions  $t = 1, \dots, T-1$  (the range on which  $\xi_t$  is defined, (17)) yields

$$Q(\theta | \theta^{(n)}) = \underbrace{\sum_i \gamma_1(i) \log \pi_i}_{\text{initial}} + \underbrace{\sum_{t=1}^{T-1} \sum_{i,j} \xi_t(i, j) \log A_{ij}}_{\text{transitions}} + \underbrace{\sum_{t=1}^T \sum_i \gamma_t(i) \log b_i(r_t)}_{\text{emissions}}. \quad (26)$$

Crucially, the three groups of parameters  $(\boldsymbol{\pi})$ ,  $(A)$ ,  $(\mu, \sigma^2)$  appear in *separate* additive terms, so the M-step decouples into three independent maximisations.

## 4.3 The M-step updates

**Initial distribution.** Maximise  $\sum_i \gamma_1(i) \log \pi_i$  subject to  $\sum_i \pi_i = 1$ . With Lagrange multiplier  $\lambda$ , the stationarity condition  $\partial_{\pi_i} [\sum_i \gamma_1(i) \log \pi_i + \lambda(1 - \sum_i \pi_i)] = 0$  gives  $\gamma_1(i)/\pi_i = \lambda$ , hence  $\pi_i = \gamma_1(i)/\lambda$  with  $\lambda = \sum_k \gamma_1(k)$  fixed by the constraint. Since  $\gamma_1(\cdot)$  is itself a probability distribution over the states at  $t = 1$  (Proposition 3),  $\sum_k \gamma_1(k) = 1$ , so  $\lambda = 1$  and

$$\boxed{\pi_i^{(n+1)} = \gamma_1(i).} \quad (27)$$

(The implementation divides by  $\sum_k \gamma_1(k)$  explicitly rather than relying on the exact identity, which is more robust to floating-point error; the two agree because the sum is 1.)

**Transition matrix.** For each row  $i$ , maximise  $\sum_j (\sum_t \xi_t(i, j)) \log A_{ij}$  subject to  $\sum_j A_{ij} = 1$ . The identical Lagrange argument gives  $A_{ij} \propto \sum_t \xi_t(i, j)$ , and using  $\sum_j \xi_t(i, j) = \gamma_t(i)$  to normalise,

$$\boxed{A_{ij}^{(n+1)} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \quad (28)$$

The numerator is the expected number of  $i \rightarrow j$  transitions; the denominator is the expected number of visits to  $i$  from which a transition occurs — the estimator is exactly an “expected count” ratio.

**Gaussian emissions.** Only the emission term of (26) involves  $(\mu_i, \sigma_i^2)$ :

$$Q_i(\mu_i, \sigma_i^2) = \sum_{t=1}^T \gamma_t(i) \left( -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma_i^2 - \frac{(r_t - \mu_i)^2}{2\sigma_i^2} \right).$$

Setting  $\partial Q_i / \partial \mu_i = 0$ :  $\sum_t \gamma_t(i)(r_t - \mu_i) / \sigma_i^2 = 0 \Rightarrow$

$$\boxed{\mu_i^{(n+1)} = \frac{\sum_t \gamma_t(i) r_t}{\sum_t \gamma_t(i)}. \quad (29)$$

Setting  $\partial Q_i / \partial \sigma_i^2 = 0$ :  $\sum_t \gamma_t(i) \left( -\frac{1}{2\sigma_i^2} + \frac{(r_t - \mu_i)^2}{2\sigma_i^4} \right) = 0 \Rightarrow$

$$\boxed{(\sigma_i^2)^{(n+1)} = \frac{\sum_t \gamma_t(i) (r_t - \mu_i^{(n+1)})^2}{\sum_t \gamma_t(i)}}. \quad (30)$$

These are exactly *responsibility-weighted* sample mean and variance, with the soft assignment  $\gamma_t(i)$  playing the role of a fractional membership of observation  $t$  in state  $i$ . Writing  $N_i := \sum_t \gamma_t(i)$  for the expected occupancy of state  $i$ , the updates are  $\mu_i = N_i^{-1} \sum_t \gamma_t(i) r_t$  and  $\sigma_i^2 = N_i^{-1} \sum_t \gamma_t(i) (r_t - \mu_i)^2$ .

#### 4.4 The algorithm

**Baum–Welch (one EM sweep).** Given  $\theta^{(n)}$ :

1. **E-step.** Run the forward (1) and backward (2) recursions in log-space; form  $\gamma_t(i)$  (3) and  $\xi_t(i, j)$  (4); record  $\log L(\theta^{(n)})$ .
2. **M-step.** Update  $\pi, A, \mu, \sigma^2$  by (27)–(30); apply the variance floor and any sign/zero constraints (Section 5).
3. **Convergence.** Stop when  $\log L(\theta^{(n+1)}) - \log L(\theta^{(n)}) < \text{tol}$  or the iteration cap is reached.

One sweep costs  $O(K^2T)$ , dominated by the forward–backward pass.

### 5 Modelling decisions and their justification

The bare EM recursions above are textbook. What makes the estimator usable on real return data is a set of deliberate choices, each addressing a concrete failure mode.

#### 5.1 Why three Gaussian states

Three states is the smallest number that can encode the qualitative trichotomy (*down / sideways / up*) that practitioners reason about, while keeping the parameter count low ( $K^2 - K = 6$  free transition probabilities,  $2K = 6$  emission parameters,  $K - 1 = 2$  initial probabilities). A Gaussian emission is the maximum-entropy choice given a state-specific mean and variance, and it makes the M-step closed-form. Its known weakness — thin tails relative to real returns — is discussed in Section 10.

#### 5.2 The variance floor: avoiding the likelihood singularity

The Gaussian-emission likelihood is *unbounded above*. If a state’s mean  $\mu_i$  coincides with a single observation  $r_\tau$  and its variance  $\sigma_i^2 \rightarrow 0$ , then  $b_i(r_\tau) = (2\pi\sigma_i^2)^{-1/2} \rightarrow \infty$ . The term of the path-sum (7) in which  $S_\tau = i$  then contains this diverging factor while all its remaining factors (the transitions and the other emissions  $b_{S_u}(r_u)$ ,  $u \neq \tau$ ) stay bounded below by a positive constant as  $\sigma_i^2 \rightarrow 0$ ; hence that single term, and with it  $L(\theta)$ , diverges to  $+\infty$ . EM can march toward such a degenerate spike, “explaining” one day perfectly and collapsing the state. We forbid this by flooring the variance after each M-step,

$$\sigma_i^2 \leftarrow \max(\sigma_i^2, \varepsilon), \quad \varepsilon = \text{var\_floor} (= 10^{-8} \text{ by default}), \quad (31)$$

which keeps every emission density bounded and the optimisation well-posed.

### 5.3 Non-convexity and multiple restarts

The log-likelihood surface is multimodal; EM converges only to a local maximum (Proposition 5), and which one depends on the initial  $\theta^{(0)}$ . We therefore run `n_restarts` independent fits from randomised initialisations and keep the one with the highest log  $L$ . Means are initialised by spreading them across empirical return quantiles (so distinct restarts explore distinct mode assignments), variances around the sample variance, and the transition matrix is initialised *sticky*.

### 5.4 Sticky transition initialisation

Financial regimes persist for weeks or months, not single days. We initialise  $A$  with large diagonal mass,

$$A_{ii}^{(0)} = 0.9, \quad A_{ij}^{(0)} = \frac{0.1}{K-1} \quad (j \neq i), \quad (32)$$

encoding the prior belief that “tomorrow’s regime is probably today’s”. This both speeds convergence and steers EM away from implausible rapidly-switching solutions.

### 5.5 Identifiability: breaking label-switching by ordering

The likelihood (7) is *invariant* under any permutation of the state labels: relabelling states and permuting  $(\pi, A, \mu, \sigma^2)$  accordingly leaves  $L$  unchanged. Hence there are  $K!$  equivalent global modes, and “state 1” from one fit need not be the bear state. We remove this ambiguity by adopting the canonical ordering

$$\mu_1 \leq \mu_2 \leq \mu_3, \quad (33)$$

i.e. after fitting we sort the states by mean ascending and permute all parameters (and  $A$  by the corresponding symmetric reindexing  $A \mapsto A[\sigma, \sigma]$ ). Index 1 is then always *bear* (lowest mean), 2 *neutral*, 3 *bull*. This is what lets downstream code refer to `P(bull)` unambiguously.

The ordering breaks the symmetry *uniquely* precisely when the means are distinct,  $\mu_1 < \mu_2 < \mu_3$ , which holds almost surely for continuous return data. On the measure-zero event of an exact tie  $\mu_i = \mu_j$  (with  $\sigma_i^2 \neq \sigma_j^2$ ) the ordering by mean alone is not unique; the implementation then breaks the residual tie by original index (`np.argsort`), a deterministic but arbitrary convention. A fully canonical rule would sort lexicographically on  $(\mu_k, \sigma_k^2)$ .

### 5.6 Optional economic constraints

Two optional constraints inject domain knowledge directly into the M-step:

- **Pinned neutral mean** (`pin_neutral_zero`): force  $\mu_{\text{neutral}} = 0$  after (29), encoding “the sideways regime has no drift”. This is a hard equality constraint on one coordinate; the remaining updates are unchanged.
- **Sign constraint** (`enforce_sign`): project the extreme means onto the economically admissible half-lines,  $\mu_{\text{bear}} \leftarrow \min(\mu_{\text{bear}}, 0)$  and  $\mu_{\text{bull}} \leftarrow \max(\mu_{\text{bull}}, 0)$ , so the bear (bull) regime cannot end up with a positive (negative) drift.

Both act on the mean coordinate only, after the unconstrained update. The monotonicity guarantee of Proposition 5 is stated for the *unconstrained* M-step, so it does not transfer automatically under a projection; what rescues it here is concavity. Each emission objective  $Q_i(\mu_i, \sigma_i^2)$  is, as a function of  $\mu_i$ , a negative quadratic and hence strictly concave with unconstrained maximiser (29). For a strictly concave function on an interval, the maximiser over a closed feasible subinterval is the

point of that subinterval nearest the unconstrained maximiser — exactly what the clip  $\min(\cdot, 0) / \max(\cdot, 0)$  (or the equality  $\mu_{\text{neutral}} = 0$ ) computes. The projected value is therefore the *constrained* maximiser of  $Q$  over the feasible set, so the M-step still maximises  $Q$  subject to the constraint and the monotone-ascent guarantee survives. (The sign clip is applied to the current lowest- and highest-mean states, which under our ordering are the bear and bull states.)

## 6 Decoding: the Viterbi algorithm

Whereas  $\gamma_t$  gives the marginally most-likely state at each  $t$  *separately*, decoding asks for the single most-likely *path*

$$s^* = \arg \max_{s_{1:T}} P(S_{1:T} = s_{1:T} \mid r_{1:T}, \theta) = \arg \max_{s_{1:T}} P(S_{1:T} = s_{1:T}, r_{1:T} \mid \theta), \quad (34)$$

the second equality holding because the data term  $P(r_{1:T})$  does not depend on the path. The Viterbi algorithm solves this by the same dynamic-programming factorisation, with  $\max$  in place of  $\sum$ .

**Definition 6** (Viterbi variable).

$$\delta_t(i) := \max_{s_{1:t-1}} P(S_{1:t-1} = s_{1:t-1}, S_t = i, r_{1:t} \mid \theta), \quad (35)$$

the probability of the best path *ending* in state  $i$  at time  $t$ .

**Proposition 6** (Viterbi recursion).

$$\delta_1(i) = \pi_i b_i(r_1), \quad (36)$$

$$\delta_t(j) = \left( \max_i \delta_{t-1}(i) A_{ij} \right) b_j(r_t), \quad \psi_t(j) = \arg \max_i \delta_{t-1}(i) A_{ij}, \quad (37)$$

with the optimal path recovered by back-tracking from  $s_T^* = \arg \max_i \delta_T(i)$  via  $s_t^* = \psi_{t+1}(s_{t+1}^*)$ .

*Proof.* The recursion is Bellman’s principle of optimality. By the joint factorisation (6), the value of a path ending in  $S_t = j$  factorises as (value of its length- $(t-1)$  prefix ending in some  $S_{t-1} = i$ )  $\times A_{ij} b_j(r_t)$ , where the last two factors depend on the prefix only through its endpoint  $i$ . Hence, *exchange argument*: if an optimal length- $t$  path ending in  $j$  had a prefix that were not itself the best length- $(t-1)$  path ending in its own endpoint  $i$ , we could substitute that better prefix — leaving  $A_{ij} b_j(r_t)$  unchanged — and strictly increase the total, contradicting optimality. Therefore the optimal prefix is  $\delta_{t-1}(i)$  for the appropriate  $i$ , and maximising the product over the predecessor  $i$  gives the stated recursion;  $\psi_t(j)$  stores the  $\arg \max$  so the maximiser can be reconstructed.  $\square$

As with the forward pass, the computation is carried out on  $\log \delta$  to avoid underflow, turning products into sums and the  $\max$  into a  $\max$  of sums; this is *exact* (the logarithm is monotone, so it commutes with  $\max$ ). Viterbi is used for offline analysis — e.g. painting the historical regime timeline — not for live trading, which must be causal.

## 7 Online inference: filtering and prediction

For trading we may use *only past and present* data at each decision point; using  $\gamma_t$  (which conditions on the whole series, including the future) would constitute look-ahead. The causal analogue of  $\gamma_t$  is the *filtered belief*.

**Definition 7** (Filtered belief).

$$\hat{\gamma}_t(i) := P(S_t = i \mid r_{1:t}, \theta) = \frac{\alpha_t(i)}{\sum_j \alpha_t(j)}. \quad (38)$$

This is just the forward variable normalised across states at time  $t$  — it uses no backward pass and hence no future information. In log-space the code computes  $\log \hat{\gamma}_t(i) = \ell \alpha_t(i) - \text{LSE}_j \ell \alpha_t(j)$ .

**One-step state prediction.** Given the belief at  $t$ , the predicted state distribution for  $t + 1$  propagates through the transition matrix. Writing the belief as a column vector  $\hat{\gamma}_t \in \mathbb{R}^K$ ,

$$p_{t+1}(j) := P(S_{t+1} = j \mid r_{1:t}) = \sum_i \hat{\gamma}_t(i) A_{ij} = (A^\top \hat{\gamma}_t)_j. \quad (39)$$

(The transpose appears because  $A$  is row-stochastic:  $A_{ij}$  maps *from*  $i$  *to*  $j$ , so summing over the originating index  $i$  is multiplication by  $A^\top$ .)

**Predictive moments of tomorrow's return.** Conditional on  $r_{1:t}$ , the next return  $r_{t+1}$  is distributed as a *mixture of Gaussians* with weights  $p_{t+1}(j)$ :

$$r_{t+1} \mid r_{1:t} \sim \sum_{j=1}^K p_{t+1}(j) \mathcal{N}(\mu_j, \sigma_j^2). \quad (40)$$

Its first two predictive moments follow from the tower property. The mean is

$$m_{t+1} := \mathbb{E}[r_{t+1} \mid r_{1:t}] = \sum_j p_{t+1}(j) \mu_j, \quad (41)$$

and, using  $\mathbb{E}[X^2] = \text{Var}(X) + \mathbb{E}[X]^2$  within each component,  $\mathbb{E}[r_{t+1}^2 \mid r_{1:t}] = \sum_j p_{t+1}(j) (\sigma_j^2 + \mu_j^2)$ , so the predictive variance is

$$v_{t+1} := \text{Var}(r_{t+1} \mid r_{1:t}) = \sum_j p_{t+1}(j) (\sigma_j^2 + \mu_j^2) - m_{t+1}^2. \quad (42)$$

Equation (42) is the *law of total variance* with the regime  $S_{t+1}$  as the conditioning variable, the relevant law being the one-step predictive  $S_{t+1} \sim p_{t+1}$ :

$$\text{Var}(r_{t+1} \mid r_{1:t}) = \mathbb{E}_{S_{t+1} \sim p_{t+1}}[\sigma_{S_{t+1}}^2] + \text{Var}_{S_{t+1} \sim p_{t+1}}(\mu_{S_{t+1}}), \quad (43)$$

i.e. the expected within-regime variance plus the across-regime dispersion of means, both taken under  $p_{t+1}$ . (Both terms are conditional on  $r_{1:t}$  through  $p_{t+1}$ ; the inner  $\mathbb{E}$  and  $\text{Var}$  average over the predicted regime.) This is the single most important output for sizing: in a confident low-volatility bull belief  $v_{t+1}$  is small, whereas a belief that mixes a high-variance bear state inflates  $v_{t+1}$  even when the directional mean  $m_{t+1}$  is ambiguous.

## 8 From belief to position

A trading rule is a map from the predictive summary  $(p_{t+1}, m_{t+1}, v_{t+1})$  to a target position  $w_{t+1} \in [-L, L]$ , where  $w = +1$  denotes fully long,  $w = -1$  fully short, and  $L$  is a leverage cap. Three rules are provided.

## 8.1 Threshold rule

The simplest rule acts only on directional confidence:

$$w_{t+1} = \begin{cases} +1, & p_{t+1}(\text{bull}) > \tau, \\ -1, & p_{t+1}(\text{bear}) > \tau, \\ 0, & \text{otherwise,} \end{cases} \quad (44)$$

for a confidence threshold  $\tau \in (\frac{1}{2}, 1)$ . It is robust and easy to reason about, but ignores the magnitude of the expected edge and the predictive variance.

## 8.2 Mean–variance (fractional-Kelly) rule

The rule that carries most of the model’s value sizes the position by the *risk-adjusted* expected return. Consider a one-period mean–variance objective for a position  $w$  in an asset with predictive mean  $m_{t+1}$  and variance  $v_{t+1}$ ,

$$U(w) = w m_{t+1} - \frac{\lambda}{2} w^2 v_{t+1}, \quad (45)$$

where  $\lambda > 0$  is a risk-aversion coefficient. Setting  $U'(w) = m_{t+1} - \lambda w v_{t+1} = 0$  gives the optimum

$$\boxed{w_{t+1}^* = \frac{m_{t+1}}{\lambda v_{t+1}}}, \quad \text{then clip to } [-L, L]. \quad (46)$$

Equation (46) is the *exact* optimum of the stated mean–variance utility  $U$ . It is closely related to — but not identical with — the Kelly criterion, and it is worth being precise about the difference, as the two are routinely conflated. The Kelly (growth-optimal) fraction maximises the expected log-growth  $\mathbb{E}[\log(1 + w r_{t+1})]$ . Two caveats apply for Gaussian returns:

- (i) **The objective is degenerate.** A Gaussian has full support on  $\mathbb{R}$ , so for any  $w \neq 0$  the event  $\{1 + w r_{t+1} < 0\}$  has positive probability and  $\mathbb{E}[\log(1 + w r_{t+1})] = -\infty$ . Literal log-growth is therefore ill-defined here; the operative criterion is the *quadratic* utility  $U$ , not log-utility.
- (ii)  **$m/v$  is an approximation.** Expanding  $\log(1 + x) \approx x - \frac{1}{2}x^2$  gives  $\mathbb{E}[\log(1 + w r_{t+1})] \approx w m_{t+1} - \frac{1}{2} w^2 (v_{t+1} + m_{t+1}^2)$ , whose maximiser is  $w^* = m_{t+1} / (v_{t+1} + m_{t+1}^2)$ . Only after the further approximation  $m_{t+1}^2 \ll v_{t+1}$  — valid for daily returns, where the squared drift is tiny next to the variance — does this reduce to  $m_{t+1} / v_{t+1}$ .

With those qualifications, (46) at  $\lambda = 1$  is the small-return approximation to full Kelly, and  $\lambda > 1$  is *fractional* Kelly (a deliberately scaled-down bet). The economically important feature is robust to all of this: the appearance of  $v_{t+1}$  in the denominator is exactly why the regime model is useful for risk control — even with an unchanged directional view  $m_{t+1}$ , the position automatically shrinks when the belief admits a high-variance (bear/stressed) regime. The clip enforces the leverage cap.

## 8.3 Smooth (tanh) rule

A continuous, bounded alternative maps the expected return through a hyperbolic tangent,

$$w_{t+1} = \tanh\left(\frac{m_{t+1}}{c}\right), \quad (47)$$

where  $c > 0$  is the return scale at which the position reaches  $\tanh(1) \approx 0.76$  of full size. It avoids the discontinuity of the threshold rule and needs no variance estimate, at the cost of not penalising risk directly.

## 8.4 Transaction-cost control: the no-trade band

Rebalancing every day to a slightly different target incurs transaction costs that can erase a thin edge. We impose hysteresis: letting  $w^{\text{prev}}$  be the position currently held and  $w^{\text{tgt}}$  the freshly computed target,

$$w_{t+1} = \begin{cases} w^{\text{prev}}, & |w^{\text{tgt}} - w^{\text{prev}}| < \text{band}, \\ w^{\text{tgt}}, & \text{otherwise.} \end{cases} \quad (48)$$

The band should be set so that a rebalance fires only when the change in target position is large enough that the expected gain exceeds the round-trip cost of trading it.

## 9 Out-of-sample evaluation: walk-forward backtesting

A backtest must never let information from the future leak into a past decision. The harness enforces this by a *rolling-window* (walk-forward) protocol. Fix a training window length  $W$  and a refit cadence  $\rho$ . For each trading day  $t > W$ :

1. **Fit (periodically).** If  $t - W$  is a multiple of  $\rho$ , refit the HMM by Baum–Welch on the trailing window  $r_{t-W:t-1}$  only. (Daily refitting adds noise and cost; monthly,  $\rho \approx 21$ , is typical.)
2. **Filter.** Compute the filtered belief  $\hat{\gamma}_{t-1}$  on the same trailing history, then the predictive  $(p_t, m_t, v_t)$  via (39)–(42).
3. **Size.** Apply the chosen rule (Section 8) and the no-trade band to obtain the position  $w_t$  *held into* day  $t$ .
4. **Realise P&L.** The net strategy return on day  $t$  is

$$\Pi_t = w_t r_t - \kappa |w_t - w_{t-1}|, \quad (49)$$

where  $\kappa$  is the per-unit-turnover transaction cost (in the same units as returns) and  $|w_t - w_{t-1}|$  is the turnover.

Because the parameters and the belief used to choose  $w_t$  depend only on data up to  $t - 1$ , while the return  $r_t$  is realised *after* the position is set, the protocol is causal by construction. The reported diagnostics — annualised return and volatility, Sharpe ratio, maximum drawdown, Calmar ratio, directional hit rate, average gross exposure, annual turnover and cumulative cost — are all computed on the out-of-sample series  $(\Pi_t)$ .

*Remark 1* (An honesty check). On structureless data, a *correct* causal backtest should show no edge (a Sharpe near zero or negative after costs). A backtest that reports a large Sharpe on noise is almost always evidence of look-ahead leakage, not of skill. The walk-forward construction above is the standard guard against that failure.

## 10 Limitations and extensions

The model is a deliberately simple baseline. The main avenues for improvement, roughly in order of expected impact:

- **Heavy tails.** Replace the Gaussian emissions by Student- $t$  emissions to capture the fat tails of daily returns; the M-step becomes an inner EM for the  $t$  scale/degrees-of-freedom but the chain machinery is unchanged.

- **Conditional heteroskedasticity.** Let each state carry a GARCH recursion (a Markov-switching GARCH) so that volatility clusters *within* a regime, not only across regimes.
- **Autoregressive emissions.** Allow  $\mathbb{E}[r_t | S_t, r_{t-1}]$  to depend on the previous return (Markov-switching AR), capturing short-horizon momentum or reversal inside a regime.
- **Duration modelling.** A plain HMM implies geometrically distributed regime durations; a hidden *semi*-Markov model lets the dwell-time distribution be estimated.
- **Model selection.** Choose  $K$  and the emission family by an information criterion (BIC/AIC) on a validation window rather than fixing  $K = 3$  a priori.
- **Multivariate / cross-asset states.** Drive a single regime chain from a vector of returns (TTF, HH, Brent jointly) with multivariate Gaussian emissions, so a regime is a market-wide rather than per-asset object.

## 11 Symbol dictionary and code map

Table 1 maps the mathematical notation to the implementation in `src/lng_desk/hmm/`.

| Symbol                         | Meaning                              | Code                                             |
|--------------------------------|--------------------------------------|--------------------------------------------------|
| $K$                            | number of hidden states ( $= 3$ )    | <code>n_states</code>                            |
| $S_t, r_t$                     | hidden state, observed return at $t$ | <code>latent / r[t]</code>                       |
| $\pi$                          | initial state distribution           | <code>HMMParams.pi</code>                        |
| $A$                            | row-stochastic transition matrix     | <code>HMMParams.A</code>                         |
| $\mu_k, \sigma_k^2$            | emission mean, variance of state $k$ | <code>HMMParams.mu, .sigma2</code>               |
| $b_k(r)$                       | Gaussian emission density            | <code>_gaussian_logpdf (log)</code>              |
| $\alpha_t(i), \beta_t(i)$      | forward / backward variables         | <code>_forward_backward</code>                   |
| $\gamma_t(i)$                  | smoothed state posterior             | <code>gamma</code>                               |
| $\xi_t(i, j)$                  | pairwise transition posterior        | <code>xi_sum</code> (summed over $t$ )           |
| LSE                            | log-sum-exp                          | <code>_logsumexp</code>                          |
| $\hat{\gamma}_t$               | filtered belief $P(S_t   r_{1:t})$   | <code>filter</code>                              |
| $p_{t+1} = A^T \hat{\gamma}_t$ | one-step state prediction            | <code>predict_next_state</code>                  |
| $m_{t+1}, v_{t+1}$             | predictive return mean / variance    | <code>predict_return_moments</code>              |
| $s^*$                          | Viterbi MAP path                     | <code>viterbi</code>                             |
| $\varepsilon$                  | variance floor                       | <code>var_floor</code>                           |
| $\lambda, L$                   | risk aversion, leverage cap          | <code>risk_aversion, leverage_cap</code>         |
| $\tau, c$                      | threshold, tanh scale                | <code>tau, scale</code>                          |
| $W, \rho, \kappa$              | train window, refit cadence, cost    | <code>train_window, refit_every, cost_bps</code> |

Table 1: Notation-to-code dictionary.

**Summary.** The pipeline is: *fit* parameters by multi-start Baum–Welch on a trailing window (Section 4), with the variance floor and label ordering of Section 5; *filter* causally to get the regime belief (Section 7); *predict* tomorrow’s return mean and variance (41)–(42); *size* the position by a risk-aware rule (46); and *evaluate* strictly out of sample (Section 9). Every step reduces to the same dynamic-programming factorisation of the Markov chain that makes the likelihood (7) computable in linear time.